

TEB 2023/ TFB 2073/ FEM 3013: ARTIFICIAL INTELLIGENCE

SEPTEMBER 2026 SEMESTER

ASSIGNMENT LAB 07

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Part A — Logical reasoning (3 marks)

Q1. Reasoning about airline messages (3 marks)
An airline uses these rules:

- If a tweet reports a cancelled or delayed flight, it is a service complaint.
- If a tweet is a service complaint, it should be prioritised for support.
- If a tweet only asks for general flight information, it is an information request.

Classify the following messages:
A. “My flight has been cancelled and nobody has contacted me.”
B. “What time does flight AB123 depart?”
a) State the category of Tweet A and Tweet B. (1 mark)

Tweet	Category	Reason
A. “My flight has been cancelled and nobody has contacted me.”	Service Complaint	It reports a cancelled flight
B. “What time does flight AB123 depart?”	Information Request	It only asks for general flight information

b) Which tweet should be prioritised? Explain the rule-based reasoning step by step. Can you conclude that Tweet B is positive or negative? Explain why. (1 mark)

- Tweet A is prioritized
- This is because if a flight is cancelled or delayed, then it is a service complaint based on rule 1 and a service complaint is prioritized for support based on rule 2
- Thus, Tweet A is prioritized over Tweet b

- No we cannot conclude if Tweet B is positive or negative
- The rules classify a tweet by its purpose not its sentiment
- Tweet B is a neutral question with no opinion words such as good, bad or angry
- Deciding positive or negative would need a sentiment rule or model which the rule set does not contain
- Even for Tweet A, the rules only say complaint not negative

c) Complete the following Python code to implement the decision rules. (1 mark)

```
def classify_message(tweet):
        tweet = tweet.lower()

        if "cancelled" in tweet or "delayed" in tweet:
                return "Service Complaint"

        elif "what time" in tweet or "departure" in tweet:
                return "Information Request"

        else:
                return "Other"

tweet_a = "My flight has been cancelled"
tweet_b = "What time does flight AB123 depart?"
```

```
print(classify_message(tweet_a))
print(classify_message(tweet_b))
```

Explain the output produced by the code.

Output:
Service Complaint
Information Request

- tweet.lower() makes matching case-insensitive
- tweet_a becomes "my flight has been cancelled". It contains "cancelled", so the first if is True and the function returns "Service Complaint". The elif and else are skipped
- tweet_b becomes "what time does flight ab123 depart?". It contains neither "cancelled" nor "delayed", so the if is False. The elif finds "what time", so it returns "Information Request"
- print() displays each returned value on its own line

Part B — Semantic network (3 marks)

Q2. Representing airline knowledge (3 marks)
Create a semantic network using the following facts:

- An Airline operates Flights.
- A Flight has a Departure Time and a Destination.
- A Passenger books a Flight.
- A Flight can be Delayed or Cancelled.
- A Delay may cause a Passenger Complaint.

a) Draw a semantic network containing the concepts and labelled relationships above. (2 marks)

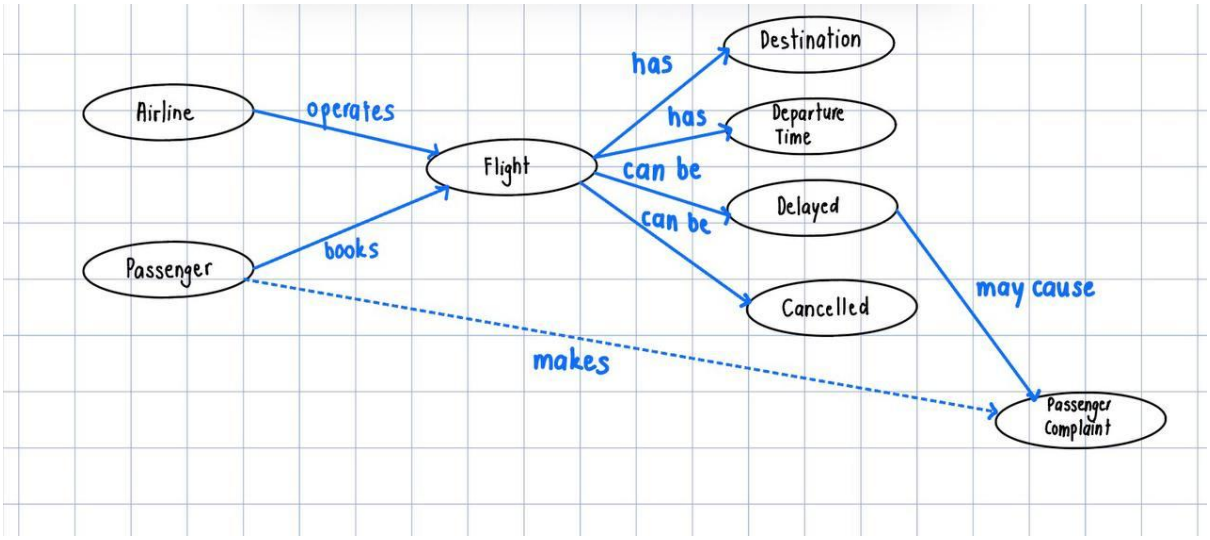


Figure 1: Semantic Network Diagram

b) Represent at least four relationships from your semantic network using Python. (1 mark)
Use the following example as a starting point:

```
semantic_network = [
    ("Airline", "operates", "Flight"),
    ("Passenger", "books", "Flight"),
    ("Flight", "has", "Destination"),
    ("Flight", "has", "Departure Time")]

for relationship in semantic_network:
    print(relationship)
```

Expected output:
('Airline', 'operates', 'Flight')
('Passenger', 'books', 'Flight')
('Flight', 'has', 'Destination')
('Flight', 'has', 'Departure Time')
Add at least one more relationship of your own.
Python Code:

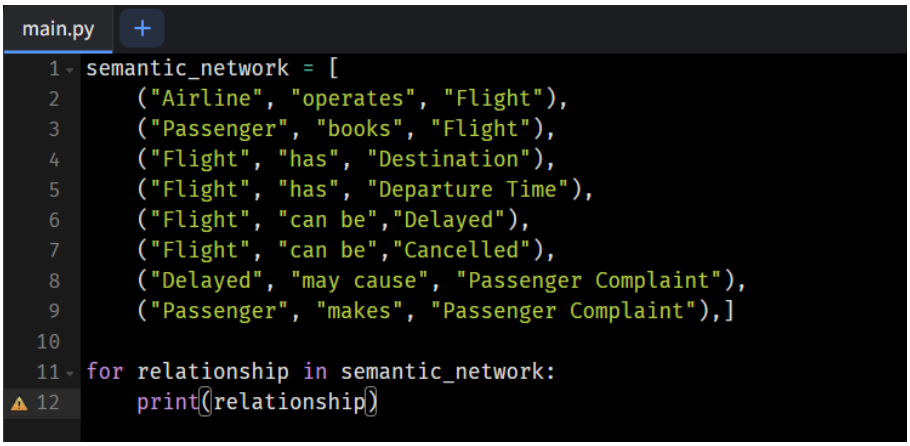


Figure 2.0: Python Code

Output:

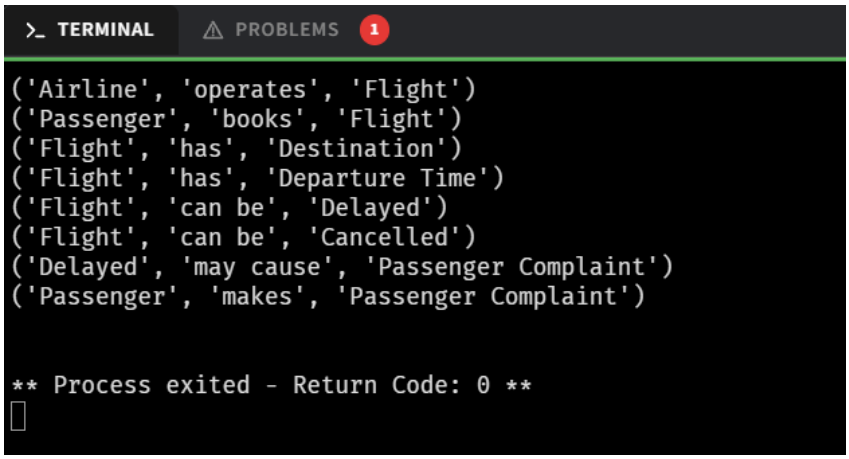


Figure 2.1: Output

Part C — Natural Language Processing (4 marks)

Q3. Text preprocessing and sentiment (2 marks)
Consider this tweet:

@SkyAir My flight is NOT good! It was delayed again. https://airline.example

a) Replace the username with USER and the URL with URL, convert the text to lowercase, and remove extra whitespace. Write the resulting text. (1 mark)

1. Replace username: USER My flight is NOT good! It was delayed again. https://airline.example
2. Replace URL: USER My flight is NOT good! It was delayed again. URL
3. Lowercase: user my flight is not good! it was delayed again. url
4. Remove extra whitespace: no extra spaces remain.
5. Results:
   user my flight is not good! it was delayed again. url

b) Complete the following Python code to perform basic text preprocessing. (1 mark)

```
import re

def preprocess_text(text):
    # Replace usernames with USER
    text = re.sub(r"@\w+", "USER", text)
    # Replace URLs with URL
    text = re.sub(r"https?:// \S+", "URL", text)
    # Convert text to lowercase
    text = text.lower()
    # Remove extra whitespace
    text = re.sub(r"\s+", " ", text).strip()
    return text

tweet = "@SkyAir My flight is NOT good! It was delayed again. https://airline.example"

clean_tweet = preprocess_text(tweet)
print(clean_tweet)
```

Write the output and explain why the word "not" should not be removed during preprocessing.

Output:
use my flight is not good! it was delayed again. url

- Not should not be removed as it's a negation word
- Which means it reverses the meaning of the sentence
- If removed, "my flight is not good" will become "my flight is good which will make it positive when it is actually begative
- Thus, the classifier will get the sentiment wrong

Q4. Evaluating a sentiment classifier (2 marks)
A classifier correctly predicts 80 out of 100 tweets, but it fails to identify most neutral tweets.

a) Is 80% accuracy alone enough to conclude that the classifier performs well? Explain. (1 mark)

- No
- Accuracy is the fraction of all tweets predicted correctly. It does not show which classes are being got right or wrong
- If the data is imbalanced, a classifier can score well by doing well on the big classes while ignoring a small one. For example, if 20 of the 100 tweets are neutral and the classifier misses all of them but gets the other 80 right, accuracy is still 80% while neutral recall is 0%
- Here the classifier fails to identify most neutral tweets, so it is not reliable across all three sentiment classes. The 80% figure hides this weakness

b) Name one metric that gives a more balanced view across negative, neutral and positive classes. Explain why it is useful. (1 mark)

- macro-averaged F1-score
- For each class (negative, neutral, positive), calculate precision and recall. The F1 is their harmonic mean: $F1 = 2 \times P \times R / (P + R)$
- Macro-F1 is the simple average of the three class F1 scores, so each class counts equally regardless of size
- It is useful here because the poor neutral performance pulls the score down sharply instead of being hidden by the large classes as in accuracy